Analysis of Variance (ANOVA)

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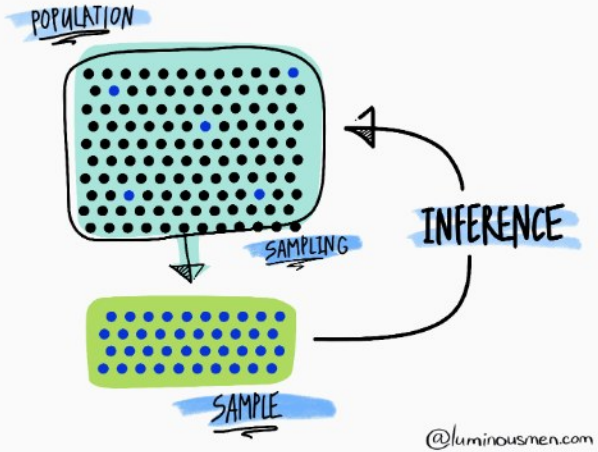
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With thanks to Luca Insolia and Federico Grazi for their help in preparing this document.

How does it work?

- The data are the outcome of a random process: they form a random sample of a (possibly conceptual) population.
- Statistical inference uses the sample to learn about the population: estimating quantities and testing hypotheses.
- Examples of research questions:
  - → pharma: does a new drug lower blood pressure more than the standard of care?
  - → earth: do magmas erupting through thick crust have a higher SiO_2 content than those erupting through thin crust?



Source: luminousmen

Case study

Short communication • Full text access Get rights and content ↗

Effects of crustal thickness on magmatic differentiation in subduction zone volcanism: A global study

Michael J. Farner ✉, Cin-Ty A. Lee Show more

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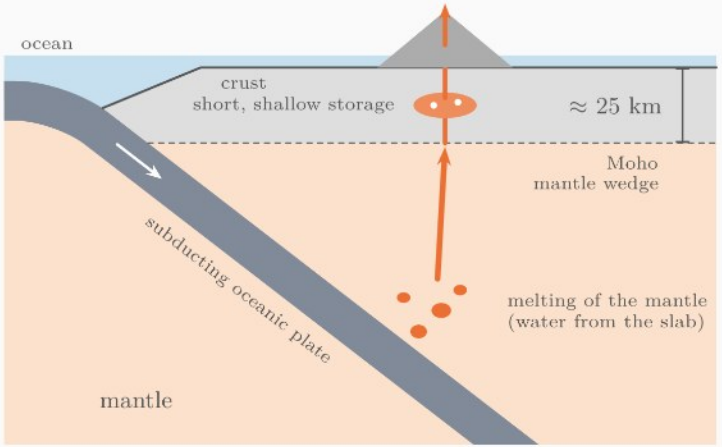
10.1016/j.epsl.2017.04.025 ↗

Hypothesis: thicker crust above subduction zones should produce lavas with higher silica (SiO_2) content.

👋 Farner, M.J. & Lee, C.-T.A. (2017). *Earth and Planetary Science Letters*, 470, 96–107. doi:10.1016/j.epsl.2017.04.025

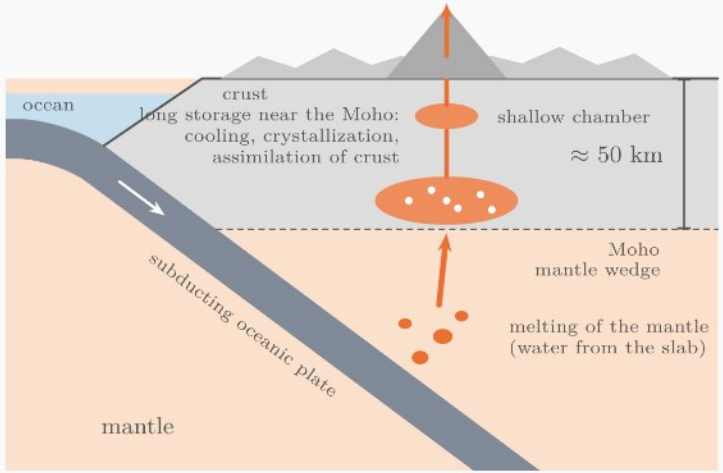
Case study — why would it be so?

Thin crust (island arc)



short path, little differentiation
⇒ low SiO_2 (basalt, basaltic andesite)

Thick crust (continental arc)



long path, extensive differentiation
⇒ high SiO_2 (andesite, dacite)

- Under **thick crust**, rising magmas travel further and are stored longer at depth: they cool, crystallize (and assimilate crust) more, so the erupted magma is more differentiated, i.e. richer in SiO_2.
- **Scientific hypothesis:** magmas erupting through thick crust have a higher SiO_2 content than those erupting through thin crust.
- **Question for this class:** is this visible in the data, and how sure can we be of what we see?

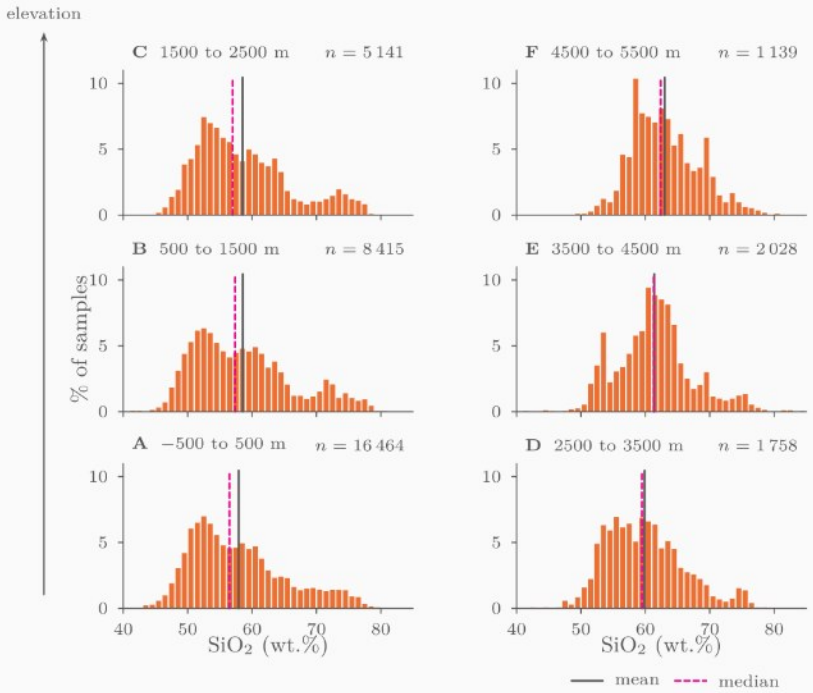
Case study — a global study



- Compare magma geochemistry of samples from multiple locations: 26 arcs, 8498 cells of 10×10 km (colours: crustal-thickness class, next slide).
- Data sourced from **GEOROC**, the public online geochemistry database.

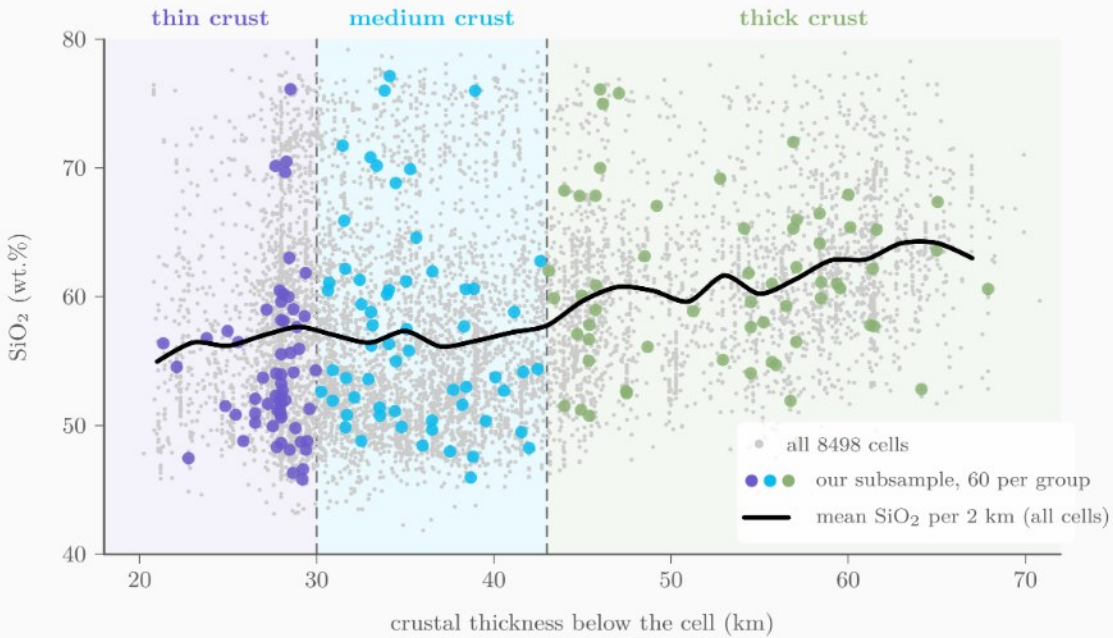
Case study — defining the groups

- Sample elevation used as a proxy for crustal thickness.
- Continuous variable (elevation, m a.s.l.) broken into discrete groups.
- Major-element concentrations can be compared between each group to test for statistically significant differences.
- E.g. SiO_2, MgO, FeO, CaO.
- In this class: three crustal-thickness groups, **thin** < 30 km, **medium** $30–43$ km, **thick** ≥ 43 km, and SiO_2.



SiO_2 by elevation bin: Fig. 2 of Farner & Lee (2017), redrawn from their data.

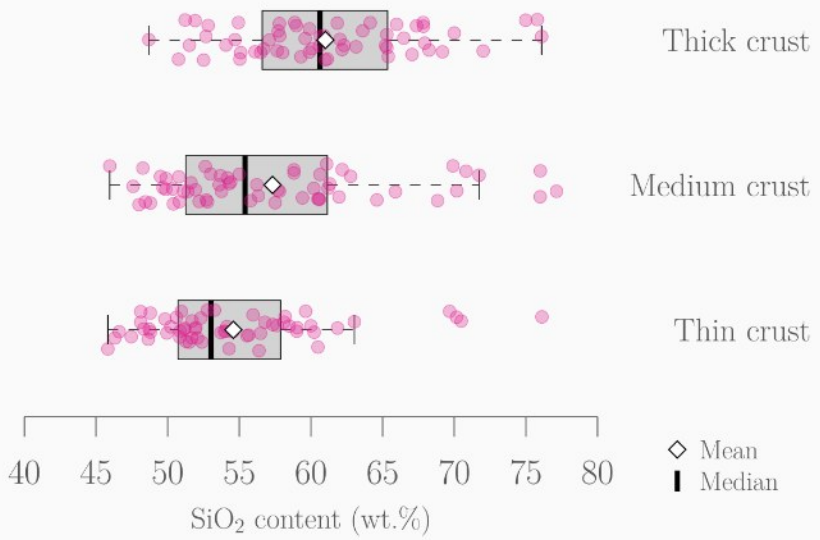
Case study — the data



- Each point is a cell. The mean SiO_2 per 2 km of thickness (black line) rises from about 56 to 63 wt.%: the trend is there, but so is a huge scatter.
- We work with one random subsample of 60 cells per group (coloured points; fixed seed), the size of a typical field study. The full dataset is on GitHub.

Case study — a first look

Group	n	mean	sd	median
thin	60	54.6	6.2	53.0
medium	60	57.3	7.7	55.4
thick	60	61.0	6.4	60.6

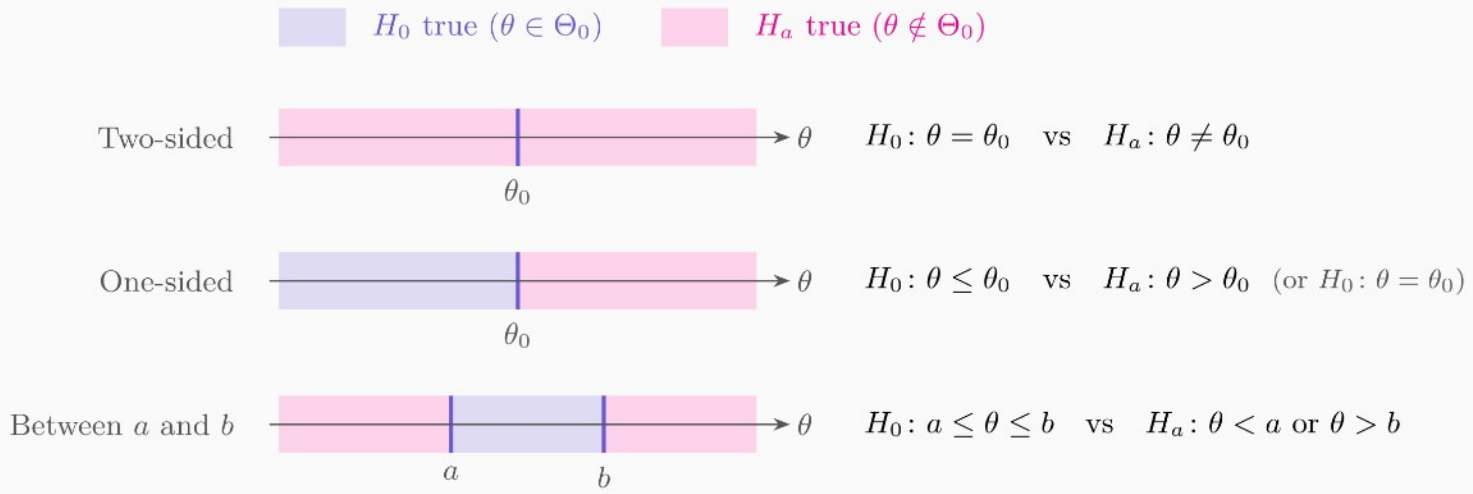


Hypothesis testing

- A scientific hypothesis is translated into two mutually exclusive statistical hypotheses:

$$H_0 : \theta \in \Theta_0 \quad \text{and} \quad H_a : \theta \notin \Theta_0$$

- Null hypothesis H_0: the statement being tested; the test weighs the evidence against it.
- Alternative hypothesis H_a: what we suspect (or hope) to be true instead.



Hypothesis testing

Statistical Hypotheses

$H_0 : \mu_{\text{thin}} = \mu_{\text{thick}}$

$H_a : \mu_{\text{thin}} < \mu_{\text{thick}}$

Sample



Data: SiO_2 content

$\left(x_{\text{tn},1}; x_{\text{tn},2}; \ldots; x_{\text{tn},n_{\text{tn}}}\right)$

$\left(x_{\text{tc},1}; x_{\text{tc},2}; \ldots; x_{\text{tc},n_{\text{tc}}}\right)$

Point estimation

$\widehat{\mu}_{\text{thin}} - \widehat{\mu}_{\text{thick}}$

Inference: Under H0

$$T_{obs} = \frac{\widehat{\mu}_{\text{thin}} - \widehat{\mu}_{\text{thick}}}{s_p\sqrt{\frac{1}{n_{\text{thin}}} + \frac{1}{n_{\text{thick}}}}} \sim St_{n_{\text{thin}}+n_{\text{thick}}-2}$$

Idea:

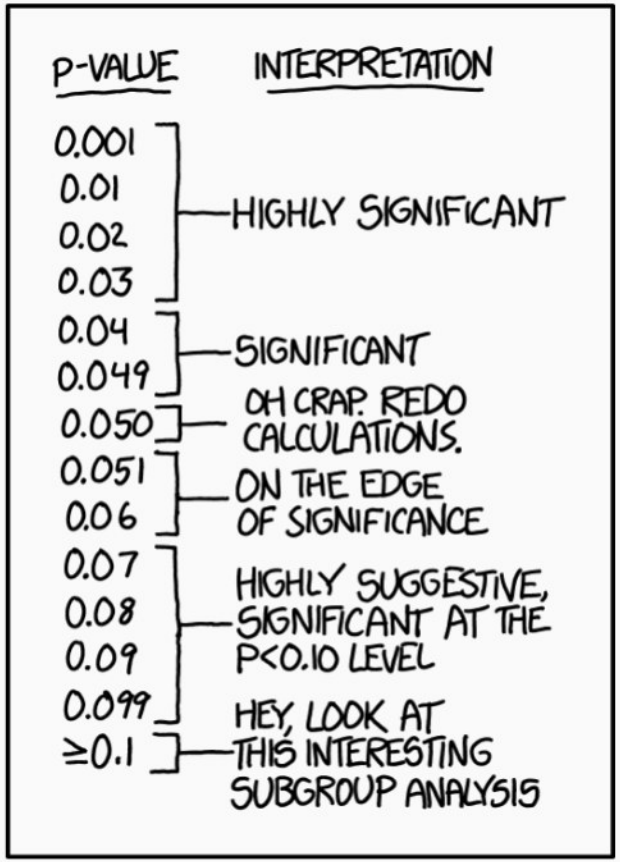
Magmas erupting through thick crust are more differentiated

Standard procedure

1. Hypotheses: state H_0 and H_a from the research question.
2. Significance level: choose α, the probability of rejecting H_0 when it is true (typically 5%), *before* looking at the results.
3. p-value: choose a suitable test (look at the data) and compute its p-value, the probability, if H_0 were true, of a result at least as extreme as the one observed (⚠️ it is **not** the probability that H_0 is true).
4. Decision:
   - $\textbf{p-value} < \alpha$: reject H_0 and accept H_a, the result is significant.
   - $\textbf{p-value} \geq \alpha$: do not reject H_0, nothing can be concluded ("absence of evidence is not evidence of absence").

Decision	H_0 is true	H_0 is false
Don't reject H_0	✅ correct (prob $1-\alpha$)	⚠️ Type II error (prob β)
Reject H_0	⚠️ Type I error (prob α)	✅ correct (prob $1-\beta$, the power)

How to understand p-values?



👋 If you want to know more have a look here.

P-values may be controversial

P-values have been misused many times because understanding what they mean is not intuitive!



▶ Click to watch the video *p-value* on YouTube (opens in a new tab).

👋 If you want to know more have a look here.

Student's t-test: a brewery story





William Sealy Gosset, "Student" (1876–1937), Guinness brewery, Dublin, 1908. Photo: public domain (Wikimedia Commons).

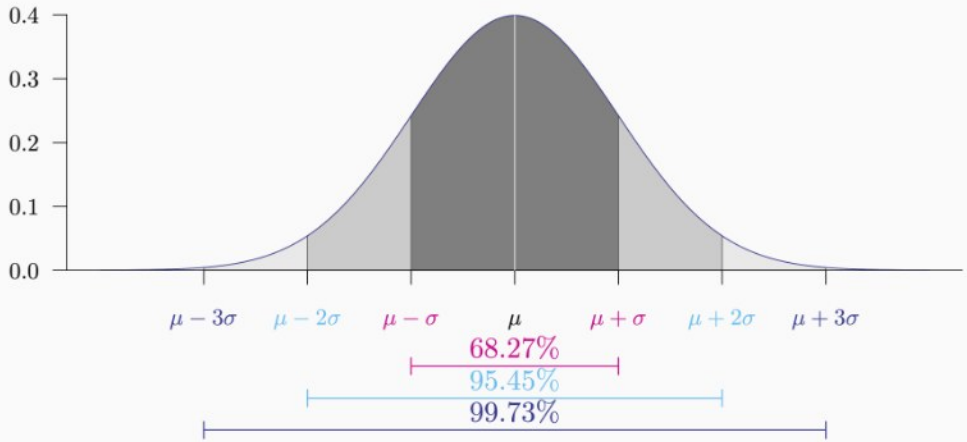
Quick review: Normal distribution

$$Y \sim N(\mu, \sigma^2), \quad f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

$$\mathbb{E}[Y] = \mu, \qquad \mathrm{Var}[Y] = \sigma^2,$$

$$Z = \frac{Y-\mu}{\sigma} \sim N(0,1), \quad f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}.$$

Probability density function of a normal distribution:



Two-sample location tests

Goal: compare the means (or locations) of two independent samples. For example:

1. A volcanologist: mean SiO_2 content of magmas erupting through thin vs medium crust.
2. An ecologist: weight of salamanders in an old-growth forest vs a clear-cut area.
3. A limnologist: duration of lake ice cover before vs after 1950.
4. Does coffee raise blood pressure? ☕
5. Does the US economy grow faster under Democratic or Republican presidents? 🇺🇸 (data at the end)
6. Is a glass of red wine as good as an hour at the gym? 🍷

We will see three tests:

1. Student's t-test: assumes equal variances.
2. Welch's t-test: no equal-variance assumption.
3. Mann-Whitney-Wilcoxon test: based on ranks, robust to outliers.

Student's t-test

Model for groups A and B ($g = A, B$ and $i = 1, \dots, n_g$), with the same variance in both groups:

$$X_{i(g)} = \mu_g + \varepsilon_{i(g)}, \qquad \varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma^2).$$

Hypotheses (usually $\mu_0 = 0$) and test statistic under H_0 (s_p = pooled standard deviation):

$$H_0 : \mu_A - \mu_B = \mu_0 \quad \text{vs} \quad H_a : \mu_A - \mu_B \left[> \text{ or } < \text{ or } \neq \right] \mu_0, \qquad T = \frac{(\overline{X}_A - \overline{X}_B) - \mu_0}{s_p \sqrt{n_A^{-1} + n_B^{-1}}} \overset{H_0}{\sim} \text{Student}(n_A + n_B - 2).$$

```
from scipy import stats
stats.ttest_ind(a = ..., b = ..., alternative = ..., equal_var = True)
```

- Needs approximately normal data without outliers (for small or moderate n).
- The equal-variance assumption is hard to verify: **we recommend avoiding this test in practice**.

Welch's t-test

Same model and hypotheses as Student's test, but each group has its own variance: $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma_g^2)$.

Test statistic under H_0 (df is computed from the data; the details are beyond the scope of this class):

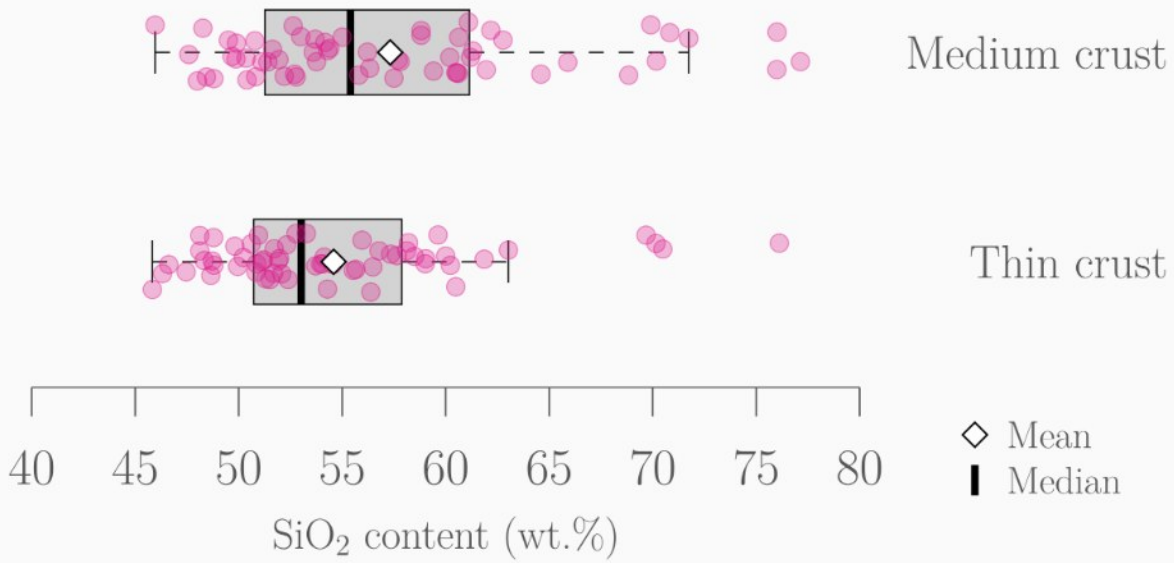
$$T = \frac{(\overline{X}_A - \overline{X}_B) - \mu_0}{\sqrt{s_A^2/n_A + s_B^2/n_B}} \overset{H_0}{\sim} \text{Student}(df).$$

```
from scipy import stats
stats.ttest_ind(a = ..., b = ..., alternative = ..., equal_var = False)
```

- Needs approximately normal data without outliers (for small or moderate n).
- Does not require equal variances. If they happen to be equal it loses only a little power compared to Student's test: **our default choice**.

Thin vs medium crust — Graph

Do magmas erupting through **thin** and **medium** crust have the same mean SiO_2 content?



👋 Random subsample of $n = 60$ cells per group (fixed seed) from the 8498 cells of Farner & Lee (2017). The tests use the same subsample.

Thin vs medium crust — Import the data

```python
# Import data
import pandas as pd
crust = pd.read_csv("https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data/arc_magmas_crust_sample.csv")

# Print the first rows of the dataset
print(crust.head(5))

# Variable of interest: SiO2 content (wt.%) for each crustal-thickness class
sio2_thin   = crust["sio2"][crust["crust_class"] == "thin"]
sio2_medium = crust["sio2"][crust["crust_class"] == "medium"]
```

```
        arc  latitude  longitude  crust_km crust_class   sio2
0    Aegean   36.3700    25.4600     27.99        thin  59.63
1  Aleutian   52.0000   178.1300     27.78        thin  51.20
2  Aleutian   52.8500   173.3100     27.99        thin  58.20
3  Aleutian   55.3409  -162.1291     33.09      medium  56.22
4  Aleutian   55.3929  -161.9471     34.07      medium  56.33
```

Thin vs medium crust — Student's t-test

```
from scipy import stats

# Student's t-test (equal_var=True assumes equal variances)
t_stat, p_value = stats.ttest_ind(sio2_thin, sio2_medium, alternative="two-sided", equal_var=True)

print("t-statistic:", round(t_stat, 4))
print("p-value:", round(p_value, 4))
```

```
t-statistic: -2.1336
p-value: 0.0349
```

Thin vs medium crust — Welch's t-test

```
from scipy import stats

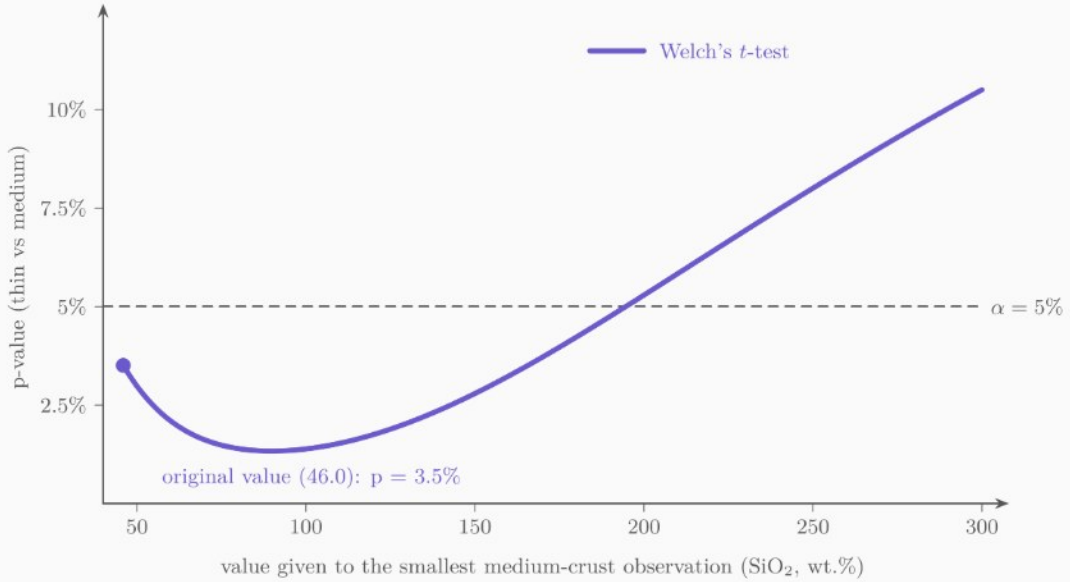
# Welch's t-test (equal_var=False makes it Welch)
t_stat, p_value = stats.ttest_ind(sio2_thin, sio2_medium, alternative="two-sided", equal_var=False)

print("t-statistic:", round(t_stat, 4))
print("p-value:", round(p_value, 4))
```

```
t-statistic: -2.1336
p-value: 0.035
```

What if one value were wrong?

The smallest medium-crust value is an Andean basalt with $SiO_2 = 46.0$ wt.%. Suppose it had been entered incorrectly: let us give it larger and larger values and redo Welch's *t*-test. (Yes, a percentage cannot exceed 100: this is a thought experiment, a typing error can produce any number.)



- At first the p-value decreases: the medium mean moves away from the thin mean.
- Then the inflated standard deviation makes the test lose significance: **5.3%** at 200, **16%** at 459.6.
- **One wrong value can flip the conclusion**: we need a test robust to such values.

Mann-Whitney-Wilcoxon test

Model for groups A and B, without normality: $X_{i(g)} = \theta_g + \varepsilon_{i(g)}$, $\varepsilon_{i(g)} \overset{iid}{\sim} (0, \sigma^2)$, where θ_g is the population location of group g.

$$H_0 : \theta_A - \theta_B = \theta_0 \quad \text{vs} \quad H_a : \theta_A - \theta_B \left[> \text{ or } < \text{ or } \neq \right] \theta_0.$$

The test statistic uses the ranks of the observations in the pooled sample, not their values ($R_{i(g)}$ = global rank of the i-th observation of group g):

$$Z = \frac{\sum_{i=1}^{n_B} R_{i(B)} - n_B(n_A + n_B + 1)/2}{\sqrt{n_A n_B (n_A + n_B + 1)/12}}.$$

```
from scipy import stats
stats.mannwhitneyu(x = ..., y = ..., alternative = ...)
```

- Robust: an extreme value only changes its rank, however large it is.
- To compare means ($\theta_g = \mu_g$) the distributions should be symmetric with equal variances; otherwise it compares locations (medians). Slightly less powerful than the t-tests when their assumptions hold.

Thin vs medium crust — Wilcoxon

```python
from scipy import stats

# Wilcoxon rank-sum test (Mann–Whitney U)
w_stat, p_value = stats.mannwhitneyu(sio2_thin, sio2_medium, alternative="two-sided")

print("Wilcoxon statistic:", round(w_stat, 4))
print("p-value:", round(p_value, 4))
```

```
Wilcoxon statistic: 1422.0
p-value: 0.0475
```

Same conclusion as Welch's *t*-test: $\mathbf{p\text{-value} = 4.75\% < 5\%}$, we reject H_0.

Robustness of the Wilcoxon test

Same experiment as before: the smallest medium-crust value is replaced by larger and larger values.



- Once the value is the largest of its group its rank no longer changes: the Wilcoxon p-value stays at **2.2%**, whatever the value (even 459.6).
- The t-test p-value crosses **5%** around 200 and reaches **16%** for the decimal slip.
- **Look at your data first**: outliers or strong skewness are a good reason to prefer the Mann-Whitney-Wilcoxon test.

Problems with multiple samples

Often we need to compare the means of more than 2 groups: for example the mean SiO_2 content of magmas erupting through **thin**, **medium** and **thick** crust. Following Farner & Lee (2017), our theory is that differentiation increases with crustal thickness:

$$\mu_{\text{thin}} < \mu_{\text{medium}} < \mu_{\text{thick}}$$

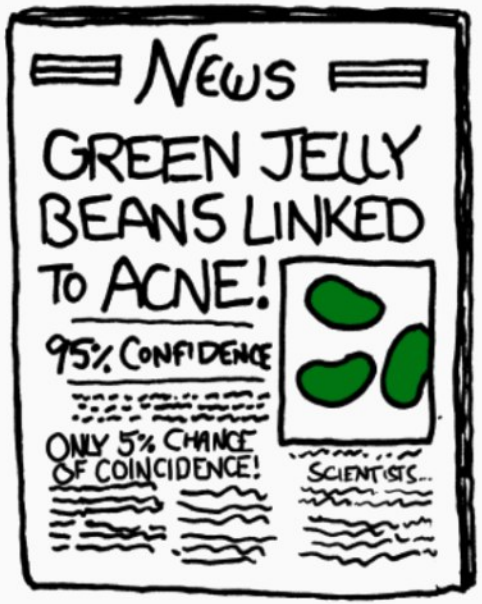
A possible approach: three one-sided tests, hoping to reject H_0 each time.

1. $H_0 : \mu_{\text{thin}} = \mu_{\text{medium}}$ vs $H_a : \mu_{\text{thin}} < \mu_{\text{medium}}$
2. $H_0 : \mu_{\text{medium}} = \mu_{\text{thick}}$ vs $H_a : \mu_{\text{medium}} < \mu_{\text{thick}}$
3. $H_0 : \mu_{\text{thin}} = \mu_{\text{thick}}$ vs $H_a : \mu_{\text{thin}} < \mu_{\text{thick}}$

But is it OK to perform several tests on the same data? 🤔

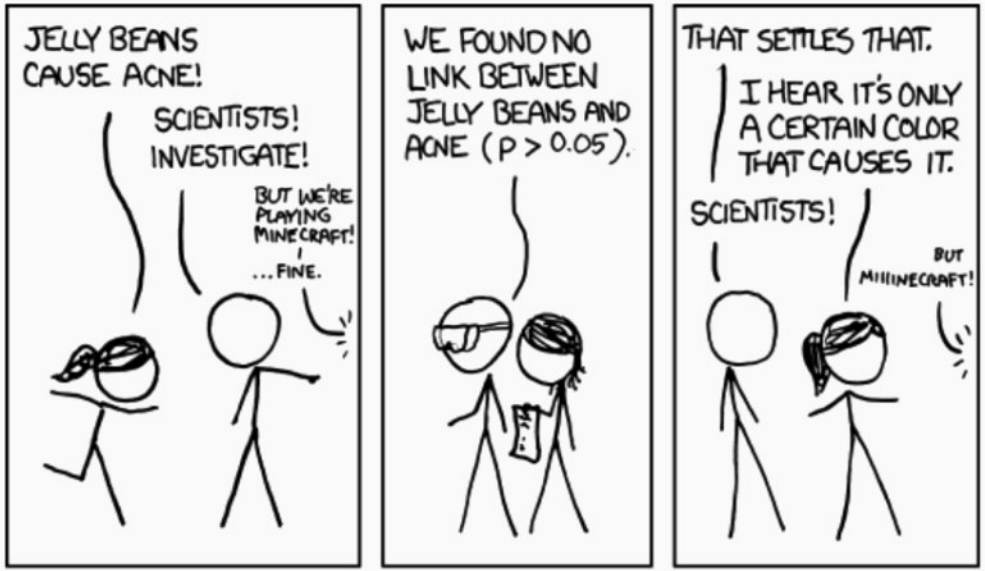
Is there a problem in doing many tests?

Are jelly beans causing acne? Maybe... but why only green ones? 😉



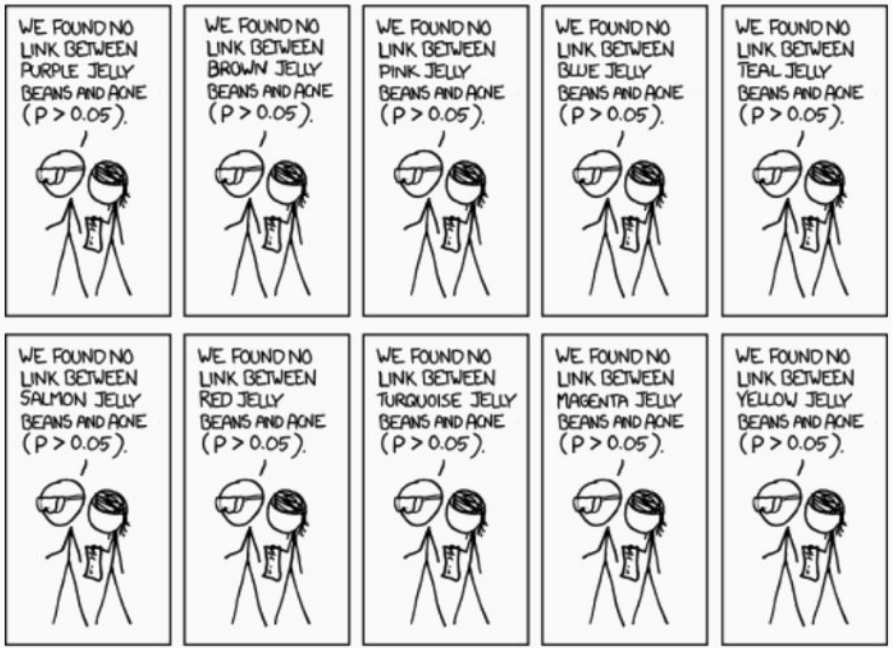
Source: xkcd

Are jelly beans causing acne?



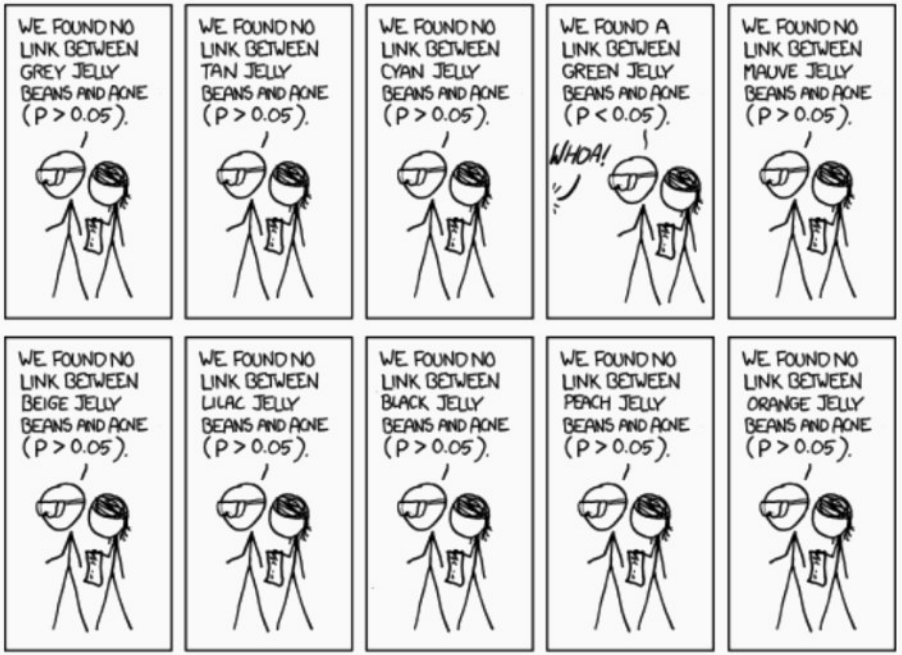
Source: xkcd

Maybe a specific color?



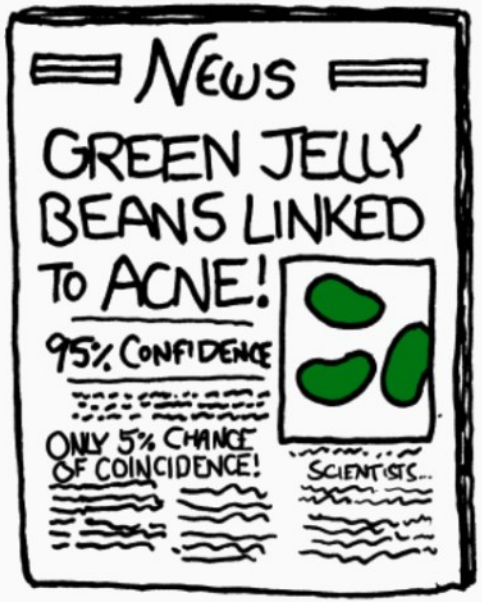
Source: xkcd

Maybe a specific color?



Source: xkcd

And finally...



Source: xkcd

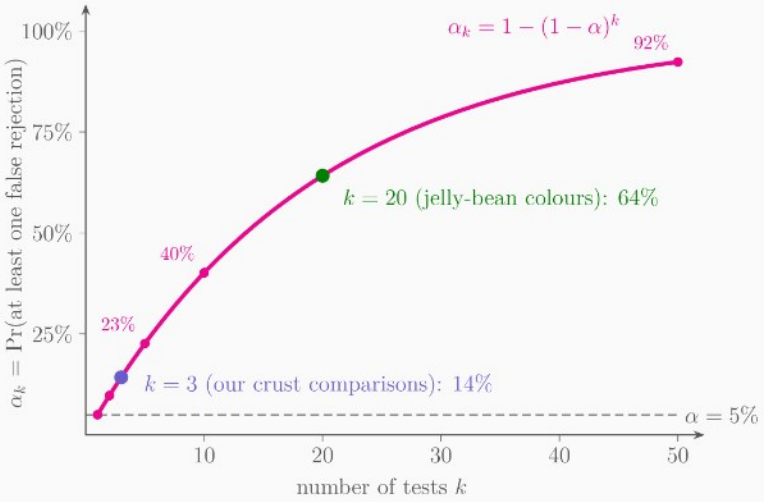
👋 If you want to know more about this comic, have a look here.

Multiple testing can be dangerous!

- A p-value is random: it depends on the data.
- One test at $\alpha = 5\%$: a 5% chance of a false alarm. k tests: k chances.
- For k independent tests, all H_0 true:

$$\alpha_k = \Pr(\text{reject at least once}) = 1 - (1 - \alpha)^k.$$

- **Jelly beans**: $k = 20$ colours, so $\alpha_{20} = 1 - 0.95^{20} \approx 64\%$. The "discovery" was expected, not lucky.
- ⚠️ Many tests, on the same or on different data, produce "significant" results even when nothing is there. Very common! 😟



Solution: a corrected level α_c

Idea: we want the chance of at least one false rejection among the k tests, α_k, to be α (say 5%). So we run each test at a smaller level α_c instead of α.

Boole's inequality: the probability of a union is at most the sum of the probabilities, whatever the dependence between the events:

$$\alpha_k = \Pr(\text{reject test 1 OR } \ldots \text{ OR reject test } k) \leq \sum_{i=1}^{k} \Pr(\text{reject test } i) = k\,\alpha_c.$$

Imposing $k\,\alpha_c = \alpha$ gives the **Bonferroni correction**:

$$\alpha_c = \frac{\alpha}{k}.$$

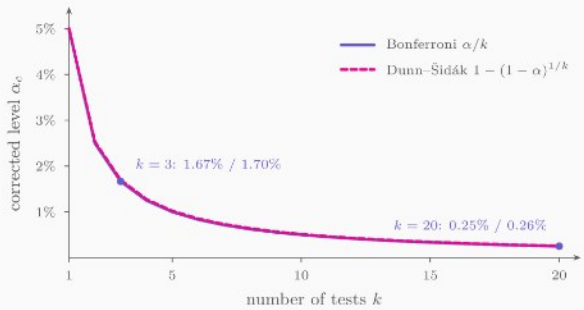
- Always valid: no assumption on the dependence between the tests, nor on how many H_0 are true.
- Our three crust comparisons: $\alpha_c = 5\%/3 \approx 1.67\%$. Jelly beans: $5\%/20 = 0.25\%$.

Bonferroni, Dunn–Šidák and beyond

- Bonferroni is conservative: α_k ends up below α, so each test loses power, more and more as k grows.
- For independent tests, α_k is known exactly: $\alpha_k = 1 - (1 - \alpha_c)^k$. Setting it to α gives the **Dunn–Šidák correction**:

$$\alpha_c = 1 - (1 - \alpha)^{1/k}.$$

k	2	3	10	20
Bonferroni	2.50%	1.67%	0.50%	0.25%
Dunn–Šidák	2.53%	1.70%	0.51%	0.26%



- Slightly less stringent, but exact only for independent tests. Pairwise comparisons sharing a group are dependent: **Bonferroni is the safe choice**.
- Thousands of tests (genomics): both corrections reject almost nothing. Methods controlling the False Discovery Rate are used instead (beyond this class).

Multiple-sample location tests

Goal: compare the means (or locations) of $G \geq 2$ independent samples. For example, the mean SiO_2 content of magmas erupting through **thin**, **medium** and **thick** crust.

Standard procedure in two steps:

- Step 1: a multiple-sample location test of H_0: "all G locations are equal". If H_0 is not rejected, we stop here.
- Step 2: otherwise, we compare the groups pairwise with two-sample tests at a corrected level α_c (Bonferroni or Dunn–Šidák) to check our scientific hypothesis.

We will see three multiple-sample location tests:

1. Fisher's one-way ANalysis Of VAriance (ANOVA): assumes equal variances.
2. Welch's one-way ANOVA: no equal-variance assumption.
3. Kruskal-Wallis test: based on ranks, robust to outliers.

👋 Step 1 is a single test, whatever the number of groups: the multiple-testing problem (and α_c) only appears at Step 2.

Fisher's one-way ANOVA

Model for G groups ($g = 1, \dots, G$ and $i = 1, \dots, n_g$), with the same variance in all groups:

$$X_{i(g)} = \mu_g + \varepsilon_{i(g)}, \qquad \varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma^2).$$

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_G \quad \text{vs} \quad H_a : \mu_i \neq \mu_j \ \text{ for at least one pair } (i, j).$$

Test statistic under H_0: the variability between the group means divided by the variability within the groups, where $N = \sum_g n_g$ and $\overline{X}$ is the mean of all N observations. Means far apart compared to the noise give a large F, i.e. a small p-value:

$$F = \frac{\frac{1}{G-1} \sum_{g=1}^{G} n_g \left(\overline{X}_g - \overline{X}\right)^2}{\frac{1}{N-G} \sum_{g=1}^{G} (n_g - 1) s_g^2} \overset{H_0}{\sim} \text{Fisher}(G-1, N-G).$$

```
from scipy import stats
stats.f_oneway(group_1, group_2, group_3)   # no "alternative": Ha is always "at least one pair differs"
```

- Needs approximately normal data without outliers (for small or moderate n_g). For $G = 2$ it is Student's test ($F = T^2$).
- The equal-variance assumption is hard to verify: **we recommend avoiding this test in practice**.

Welch's one-way ANOVA

Same model and hypotheses as Fisher's ANOVA, but each group has its own variance: $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma_g^2)$.

Test statistic under H_0: each group mean is weighted by its precision $w_g = n_g/s_g^2$ (the correction in the denominator and Δ are computed from the data; the details are beyond the scope of this class):

$$F^* = \frac{\frac{1}{G-1}\sum_{g=1}^{G} w_g \left(\overline{X}_g - \overline{X}^*\right)^2}{1 + \frac{2(G-2)}{3\Delta}} \overset{H_0}{\sim} \text{Fisher}(G-1, \Delta), \qquad \overline{X}^* = \frac{\sum_{g=1}^{G} w_g \overline{X}_g}{\sum_{g=1}^{G} w_g}.$$

For the curious: $\Delta = \left[\frac{3}{G^2-1}\sum_{g=1}^{G}\frac{1}{n_g-1}\left(1 - \frac{w_g}{\sum_{h=1}^{G} w_h}\right)^2\right]^{-1}$. For $G = 2$ it is Welch's t-test ($F^* = T^2$ and $\Delta = df$).

```
from statsmodels.stats.oneway import anova_oneway
anova_oneway(data = ..., groups = ..., use_var = "unequal", welch_correction = True)
```

- Needs approximately normal data without outliers (for small or moderate n_g).
- Does not require equal variances. If they happen to be equal it loses only a little power compared to Fisher's ANOVA: **our default choice**.

Kruskal-Wallis test

Model for G groups, without normality: $X_{i(g)} = \theta_g + \varepsilon_{i(g)}$, $\varepsilon_{i(g)} \overset{iid}{\sim} (0, \sigma^2)$, where θ_g is the population location of group g.

$$H_0 : \theta_1 = \theta_2 = \ldots = \theta_G \quad \text{vs} \quad H_a : \theta_i \neq \theta_j \ \text{ for at least one pair } (i, j).$$

The test statistic applies the ANOVA idea to the ranks of the pooled sample, not to the values: it measures how far the mean rank of each group, $\overline{R}_g$, is from the mean of all ranks, $(N+1)/2$:

$$H = \frac{12}{N(N+1)} \sum_{g=1}^{G} n_g \left(\overline{R}_g - \frac{N+1}{2} \right)^2 \overset{H_0}{\sim} \chi^2(G-1).$$

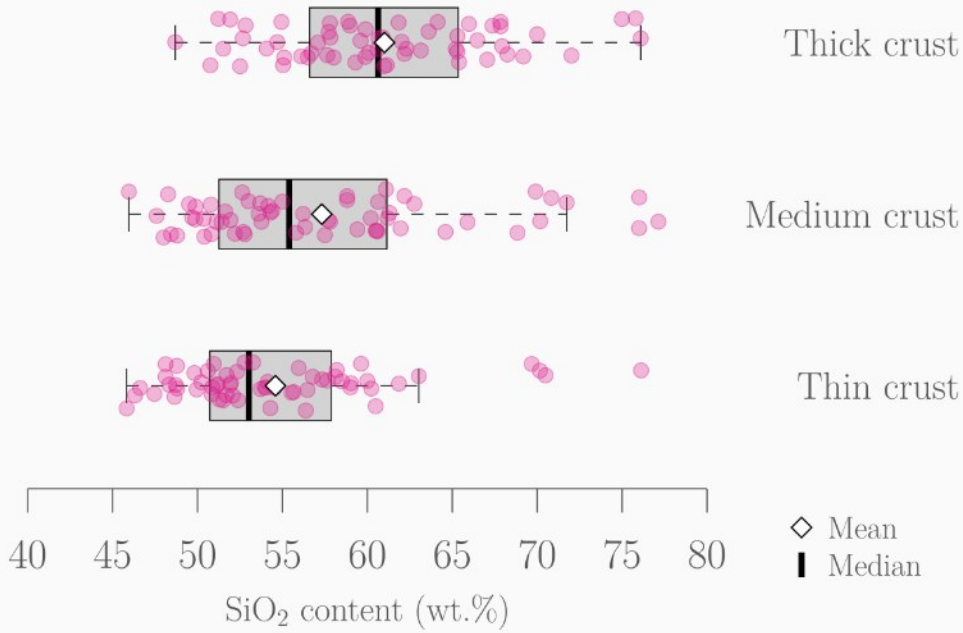
With ties, H is divided by $1 - \sum_{v=1}^{V} (t_v^3 - t_v)/(N^3 - N)$, where t_v is the number of observations sharing the v-th distinct value.

```
1 from scipy import stats
2 stats.kruskal(group_1, group_2, group_3)
```

- Robust: an extreme value only changes its rank, however large it is.
- To compare means ($\theta_g = \mu_g$) the distributions should be symmetric with equal variances; otherwise it compares locations (medians). Slightly less powerful than the ANOVAs when their assumptions hold.

Thin, medium, thick crust — Graph

Is the mean SiO_2 content of arc magmas the same for **thin**, **medium** and **thick** crust? Our theory (Farner & Lee, 2017): $\mu_{\text{thin}} < \mu_{\text{medium}} < \mu_{\text{thick}}$.



👋 Random subsample of $n = 60$ cells per group (fixed seed) from the 8498 cells of Farner & Lee (2017). The tests use the same subsample.

Thin, medium, thick crust — Import

```python
# Import data
import pandas as pd
crust = pd.read_csv("https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data/arc_magmas_crust_sample.csv")

# Variables of interest: SiO2 content (wt.%) for each crustal-thickness class
sio2_thin   = crust["sio2"][crust["crust_class"] == "thin"]
sio2_medium = crust["sio2"][crust["crust_class"] == "medium"]
sio2_thick  = crust["sio2"][crust["crust_class"] == "thick"]

# Summary by class
summary = crust.groupby("crust_class")["sio2"].agg(["count", "mean", "std", "median"])
print(summary.loc[["thin", "medium", "thick"]].round(2))
```

```
             count   mean   std  median
crust_class
thin            60  54.59  6.22   53.02
medium          60  57.32  7.72   55.40
thick           60  61.01  6.39   60.64
```

Thin, medium, thick crust — Step 1

```python
from scipy import stats
from statsmodels.stats.oneway import anova_oneway

# Fisher's one-way ANOVA
F_stat, p_fisher = stats.f_oneway(sio2_thin, sio2_medium, sio2_thick)
print("Fisher's ANOVA:  F =", round(F_stat, 3), " p-value =", f"{p_fisher:.1e}")

# Welch's one-way ANOVA
res = anova_oneway(crust["sio2"], groups=crust["crust_class"], use_var="unequal", welch_correction=True)
print("Welch's ANOVA:   F =", round(res.statistic, 3), " p-value =", f"{res.pvalue:.1e}")

# Kruskal-Wallis test
H_stat, p_kw = stats.kruskal(sio2_thin, sio2_medium, sio2_thick)
print("Kruskal-Wallis:  H =", round(H_stat, 3), " p-value =", f"{p_kw:.1e}")
```

```
Fisher's ANOVA:  F = 13.427  p-value = 3.7e-06
Welch's ANOVA:   F = 15.519  p-value = 1.1e-06
Kruskal-Wallis:  H = 29.124  p-value = 4.7e-07
```

All three p-values are far below **5%**: we reject H_0, the mean SiO_2 content is not the same in the three classes. On to Step 2.

Thin, medium, thick crust — Step 2

Three one-sided Welch t-tests (Tests 1 to 3), each at the Bonferroni level $\alpha_c = 0.05/3 \approx 1.67\%$ (Dunn–Šidák would give 1.70%).

```python
from scipy import stats

alpha_c = 0.05 / 3                                  # Bonferroni correction, 3 tests

# Test 1:  H0: mu_thin = mu_medium    vs  Ha: mu_thin < mu_medium
p_1 = stats.ttest_ind(sio2_thin, sio2_medium, alternative="less", equal_var=False).pvalue
# Test 2:  H0: mu_medium = mu_thick   vs  Ha: mu_medium < mu_thick
p_2 = stats.ttest_ind(sio2_medium, sio2_thick, alternative="less", equal_var=False).pvalue
# Test 3:  H0: mu_thin = mu_thick     vs  Ha: mu_thin < mu_thick
p_3 = stats.ttest_ind(sio2_thin, sio2_thick, alternative="less", equal_var=False).pvalue

for name, p in [("Test 1 (thin < medium) ", p_1), ("Test 2 (medium < thick)", p_2), ("Test 3 (thin < thick)  ", p_3)]:
    print(name, " p-value =", f"{p:.1e}", " -> reject H0" if p < alpha_c else " -> do not reject H0")
```

```
Test 1 (thin < medium)   p-value = 1.8e-02  -> do not reject H0
Test 2 (medium < thick)  p-value = 2.6e-03  -> reject H0
Test 3 (thin < thick)    p-value = 7.9e-08  -> reject H0
```

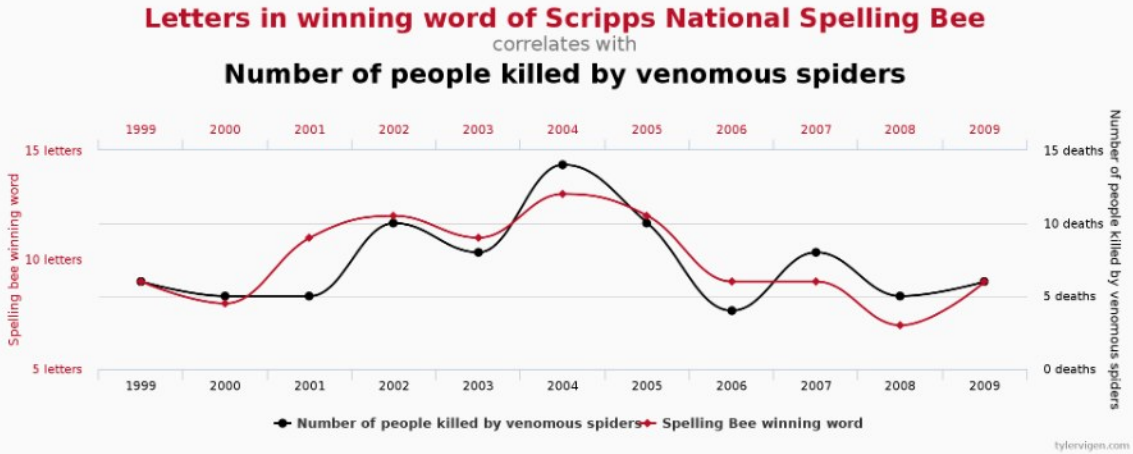
Thin, medium, thick crust — Conclusion

Test	H_a	p-value	Decision at $\alpha_c = 1.67\%$
1	$\mu_{\text{thin}} < \mu_{\text{medium}}$	1.75%	not rejected
2	$\mu_{\text{medium}} < \mu_{\text{thick}}$	0.26%	reject H_0 ✅
3	$\mu_{\text{thin}} < \mu_{\text{thick}}$	8×10^{-8}	reject H_0 ✅

- Magmas erupting through thick crust have a higher mean SiO_2 content than those erupting through medium or thin crust (Tests 2 and 3).
- Thin vs medium: p-value $= 1.75\%$, just above α_c. ⚠️ Alone at $\alpha = 5\%$ this test rejects H_0: the correction for multiple testing changes the conclusion. And not rejecting does not prove that $\mu_{\text{thin}} = \mu_{\text{medium}}$.
- The data support the thick-crust part of our theory $\mu_{\text{thin}} < \mu_{\text{medium}} < \mu_{\text{thick}}$, but not (yet) the thin-to-medium step.
- With all 8498 cells: mean SiO_2 of 57.1, 56.9 and 61.2 wt.%, p-values $< 10^{-88}$ for Tests 2 and 3 but 88% for Test 1: the thin-to-medium step is not supported. Differentiation seems only strongly enhanced once the crust is thick.

The big picture: p-hacking

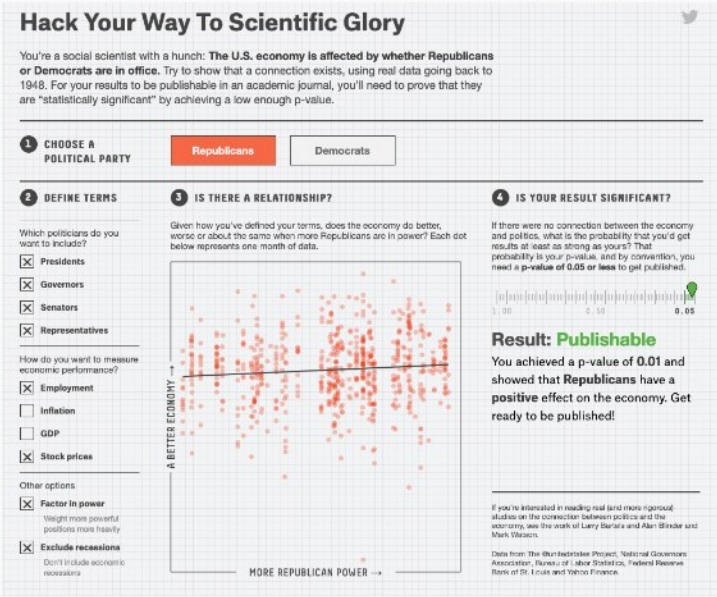
P-hacking is the misuse of data analysis to find patterns that can be presented as statistically significant: run many tests on the data and report only those that come back significant. **Example:** the number of letters in the Scripps National Spelling Bee winning word correlates with the number of people killed by venomous spiders in the US.



Source: Wikipedia on data dredging. 👋 If you want to know more, take a look here.

How to p-hack? 🤔

Back to our question: are Republicans good or bad for the US economy? **Both!!** Choose the terms, the measure of the economy and the years, and you can "prove" either. Try it **here** (a rebuild of the original FiveThirtyEight tool, archived **here**).



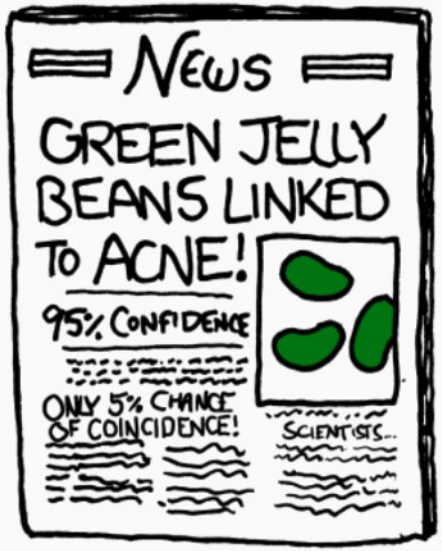
This brings us to “HARKing”

HARKing is an acronym coined by Norbert Kerr for the questionable research practice of “Hypothesizing After the Results are Known”: presenting **a post hoc** hypothesis in the introduction of a research report as if it were an **a priori** hypothesis.

How to do it? 🤔

- Test if one jelly bean colour is linked to acne.
- Pretend this was our original hypothesis.
- Et voilà!

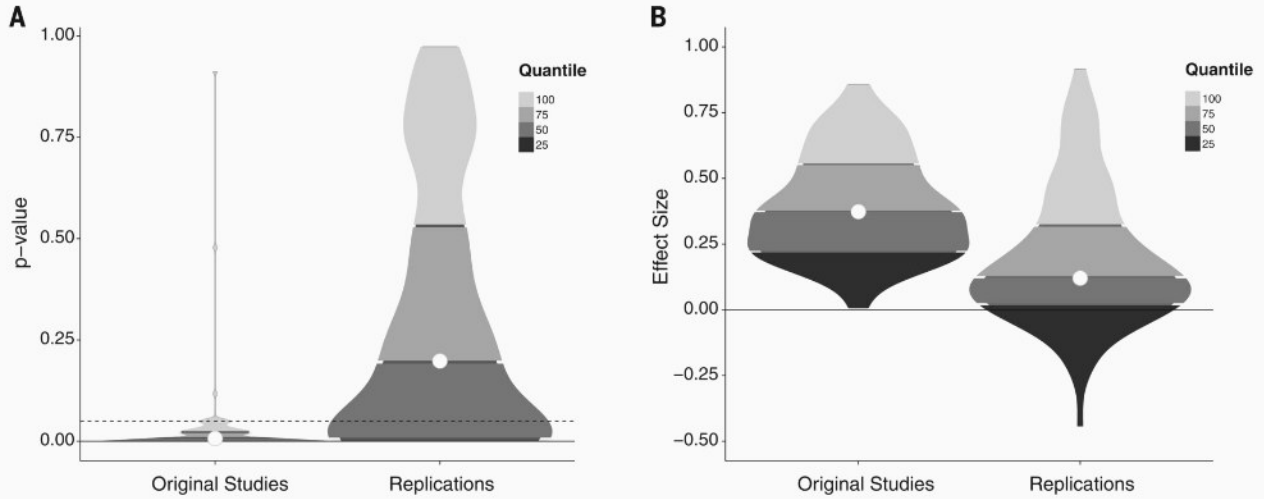
👋 If you want to know more, have a look **here** and the references therein.



Source: **xkcd**

The costs of HARKing (and p-hacking)

HARKing and p-hacking are (arguably) playing an important role in the **replication crisis in science**: when 100 published psychology studies were repeated, most p-values and effect sizes did not hold up.



Source: Open Science Collaboration, "Estimating the reproducibility of psychological science", *Science* 349 (2015). 👋 If you want to know more, take a look here.

The cost for society

Contradictory studies (coffee, wine, milk, …) make the public **cynical about science**: "find the study that sounds best to you and go with that". Once evidence can be picked like that, it has lost its power.



👋 If you want to know more, take a look **here**.

The four horsemen of irreproducibility



WORLD VIEW *A personal take on events*

ROBERT TWIG/DR

Rein in the four horsemen of irreproducibility

Dorothy Bishop *describes how threats to reproducibility, recognized but unaddressed for decades, might finally be brought under control.*

More than four decades into my scientific career, I find myself an outlier among academics of similar age and seniority: I strongly identify with the movement to make the practice of science more robust. It's not that my contemporaries are unconcerned about doing science well; it's just that many of them don't seem to recognize that there are serious problems with current practices. By contrast, I think that, in two decades, we will look back on the past 60 years — particularly in biomedical science — and marvel at how much time and money has been wasted on flawed research.

How can that be? We know how to formulate and test hypotheses in controlled experiments. We can account for unwanted variation with statistical techniques. We appreciate the need to replicate observations.

Yet many researchers persist in working in a way almost guaranteed not to deliver meaningful results. They ride with what I refer to as the four horsemen of the reproducibility apocalypse: publication bias, low statistical power, *P*-value hacking and HARKing (hypothesizing after results are known). My generation and the one before us have done little to rein these in.

In 1975, psychologist Anthony Greenwald noted that science is prejudiced against null hypotheses; we even refer to sound work supporting such conclusions as 'failed experiments'. This prejudice leads to publication bias: researchers are less likely to write up studies that show no effect, and journal editors are less likely to accept them. Consequently, no one can learn from them, and researchers waste time and resources on repeating experiments, redundantly.

That has begun to change for two reasons.

MANY RESEARCHERS PERSIST IN WORKING IN A WAY ALMOST **GUARANTEED** NOT TO DELIVER **MEANINGFUL RESULTS.**

be adequately powered. Other disciplines have yet to catch up.

I stumbled on the issue of *P*-hacking before the term existed. In the 1980s, I reviewed the literature on brain lateralization (how sides of the brain take on different functions) and developmental disorders, and I noticed that, although many studies described links between handedness and dyslexia, the definition of 'atypical handedness' changed from study to study — even within the same research group. I published a sarcastic note, including a simulation to show how easy it was to find an effect if you explored the data after collecting results (D. V. M. Bishop *J. Clin. Exp. Neuropsychol.* **12**, 812–816; 1990). I subsequently noticed similar phenomena in other fields: researchers try out many analyses but report only the ones that are 'statistically significant'.

This practice, now known as *P*-hacking, was once endemic to most branches of science that rely on *P* values to test significance of results, yet few people realized how seriously it could distort findings. That started to change in 2011, with an elegant, comic paper in which the authors crafted analyses to prove that listening to the Beatles could make undergraduates younger (J. P. Simmons *et al. Psychol. Sci.* **22**, 1359–1366; 2011). "Undisclosed flexibility," they wrote, "allows presenting anything as significant."

The term HARKing was coined in 1998 (N. L. Kerr *Pers. Soc. Psychol. Rev.* **2**, 196–217; 1998). Like *P*-hacking, it is so widespread that researchers assume it is good practice. They look at the data, pluck out a finding that looks exciting and write a paper to tell a story around this result. Of course, researchers should be free to explore their data for unexpected findings — but *P* values are meaningless when taken out of context of all the

Dorothy Bishop's four horsemen of the reproducibility apocalypse:

- publication bias
- low statistical power
- p-hacking
- HARKing

We met the last three today. The remedies she lists are now standard: pre-registration, registered reports, open data and code.

Source: D. V. M. Bishop, *Nature* 568, 435 (2019).